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Thermophysical properties of liquid Co–Cr–Mo alloys measured by electromagnetic levitation in a static magnetic field

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ABSTRACT

This study aimed to provide thermophysical property data of Co–Cr–Mo (CCM) alloys that are used as biomedical materials to understand and improve additive manufacturing processes. The density, surface tension, normal spectral emissivity, specific heat capacity and thermal conductivity of two CCM alloys that contained a low (0.053 mass%) and a high (0.251 mass%) carbon content were measured in a liquid state using an electromagnetic levitation technique. The liquidus temperatures of the CCM alloys were measured by differential scanning calorimetry. A static magnetic field was applied to the levitated CCM droplets to suppress the surface oscillation and translational motion of the droplets, and convection flow inside the droplet for each property measurement as needed. Uncertainty analysis was conducted for all thermophysical property data. No major difference was found in the thermophysical properties for the low- and high-carbon CCM melts within experimental uncertainty. An ideal solution model reproduced the CCM melt density within experimental uncertainty, however, we obtained large positive excess heat capacities of the CCM melts.

1. Introduction

Co–Cr–Mo (CCM) alloys, such as the ASTM F75, F799 and F1537 alloys, have been used as biomedical implants, such as femoral heads of joints and artificial knees, in dental applications and in hip joints [1–6] because of their excellent corrosion and wear resistance and biocompatibility. These products are manufactured by selective laser melting and electron beam melting, which are classified as powder bed fusion-type additive manufacturing (AM) technologies. Thermophysical properties are required as input parameters to model the AM process accurately [7, 8]. The addition of carbon to the CCM alloys has a significant effect on their mechanical and tribological behaviors [6, 9-11]. For example, carbon addition stabilizes the face-centered cubic matrix phase, i.e., γ −phase [10] and decreases the ε-martensite plates and stacking faults [10, 11]. Carbon (>0.0024 mass%) is incorporated into the matrix as $M_{23}C_6$-type carbides with various sizes and morphologies. These microstructural changes induce significant variations in the material properties. Fine, spherical and discontinuous island-like carbides tend to form in the grains of high-carbon alloys, which increases the strength and ductility. In contrast, coarse carbides that form at the grain boundaries embrittle the matrix and degrade the mechanical properties. Thus, the effects of carbide addition have been studied to understand the tribological properties of CCM alloys [12]. Therefore, carbon addition to CCM melts may influence their mechanical properties, and the thermophysical properties of the CCM melts with carbon addition should be investigated. However, technical problems in the thermophysical property measurements of the melt arise because of their high chemical reactivity with crucible materials and the atmosphere at high temperatures. To address these problems, the application of levitation techniques such as electrostatic levitation [13, 14] and electromagnetic levitation (EML) is proposed. Our research group has developed a high-temperature thermophysical property measurement system for liquid metals, which is termed PROSPECT [15–18]. PROSPECT consists of EML and a superconducting magnet. A superconducting magnet can suppress surface oscillation, translational motion and convection flow in the levitated droplet. Therefore, PROSPECT enables accurate density (ρ), normal spectral emissivity (ε), heat capacity at constant pressure (C_P) and thermal conductivity (κ) measurements of liquid metals. The

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magnet is not used for surface tension (σ) measurement, because the surface oscillation frequency of the droplet needs to be measured. In this study, thermophysical properties (ρ, σ, ε, C_P and κ) of two CCM alloys melts with different carbon contents: low- (LC, C content: 0.053 mass%) and high- (HC, C content: 0.251 mass%) carbon alloys were measured by PROSPECT. From these results, the influence of carbon content on the thermophysical properties of CCM melts was investigated. PROSPECT offers the advantage that thermophysical properties can be measured over a wide range of temperatures from a considerably higher temperature than the liquidus temperature to deep undercooling region without chemical contamination. However, only the liquidus temperatures of the CCM alloys were measured beforehand using differential scanning calorimetry (DSC) to calibrate the pyrometer that was used in PROSPECT.

2. Materials and measurement methods

2.1. Measurement of liquidus temperature using DSC

The chemical compositions of the starting materials of the LC and HC alloys are shown in Table 1. The Cr, Mo, Ni, Fe, Si and Mn elemental compositions were analyzed by inductively coupled plasma-optical emission spectroscopy (ICP-OES, ARCOS MV130, SPECTRO Analytical Instruments, Kleve, Germany), and the elemental C, S and O concentrations were measured by infrared detection (C and S analysis: CS-444, LECO, St. Joseph, USA; O analysis: CS-436, LECO, St. Joseph, USA). DSC analysis was performed on an STA 449 F3-TF24 Jupiter differential scanning calorimeter (NETZSCH-Gerätebau GmbH, Selb/Bayern, Germany) for the liquidus temperature measurement. The DSC instrument was calibrated by using the melting points of In, Sn, Bi, Al, Ag, Au and Ni samples in an Al_2O_3 crucible. The LC and HC samples were placed in the DSC instrument. In the DSC procedure, a heating and cooling cycle was repeated three times from 873 K to 1823 K with heating and cooling rates of 10 K/min. The DSC data for the second and third heating processes were used to determine the liquidus temperature of the sample to ensure a similar contact area between the sample and the crucible.

2.2. Measurement of thermophysical properties using prospect

The experimental apparatus and detailed procedures were previously reported [15–18], therefore, only a brief explanation is presented here. Fig. 1 shows a schematic illustration of the experimental equipment (PROSPECT). The chamber was evacuated to 10^{-3} Pa by using a turbo-molecular pump. After evacuation, Ar-5 vol% H_2 and He gasses up to normal pressure filled the chamber. The mass of the LC and HC melts was in the range of 0.72–1.03 and 0.75–1.00 g, respectively. The LC and HC samples were levitated by applying an alternating current of 380 to 570 A to the levitation coil. The thermophysical properties of the LC and HC melts were measured during levitation. A single-color pyrometer (detection wavelength range: 1.45–1.8 μm, IGA140/MB25, LumaSense Technologies, Frankfurt, Germany) measured the temperature of the levitated sample at the bottom part and was calibrated by using the liquidus temperatures of the LC and HC alloys determined independently during the DSC runs. The sample temperature was determined assuming that the normal spectral emissivity of the liquid samples that was determined at their liquidus temperature had no temperature dependence at the pyrometer detection wavelength. After the thermophysical property measurements, the chemical composition of the LC and HC alloys was measured to evaluate the chemical composition change during the thermophysical property measurements because of evaporation.

2.2.1. Density measurement

The density (ρ) was determined as follows:

$$\rho = \frac{M}{V}, \tag{1}$$

where M is the sample mass and V is the sample volume. M was determined from the average values of the mass before and after levitation. The sample was levitated under a static magnetic field of 4 T and was illuminated from a horizontal direction by a green laser at a wavelength of 532 nm to obtain the shadow image. The horizontal shadow images were taken by a high-speed camera (MC1310, Mikrotron, Unterschleißheim, Germany) equipped with an interference filter at a frame rate of 300 for 10 s for each temperature. V was determined from the average values of the volumes that were calculated from the images under the assumption that the levitated sample had a rotationally symmetrical shape around the vertical axis.

2.2.2. Surface tension measurement

The surface tension (σ) of the LC and HC melts was measured by the oscillating droplet method with EML without a static magnetic field. σ can be obtained from the surface oscillation frequency (ν_R) of the levitated droplet by Rayleigh's equation [19]:

$$\sigma = \frac{3\pi M}{8}{\nu_R}^2, \tag{2}$$

However, Rayleigh's equation can only be applied to spherical and external force-free droplets. In the case of an electromagnetically levitated droplet under gravity, the surface oscillation frequency shifts and splits due to the deformation of the levitated droplet by gravity and the electromagnetic force. In 1991, Cummings and Blackburn [20] suggested a compensation formula to measureσ using EML under gravity as follows:

$$\sigma = \frac{3\pi M}{8}\left\{\frac{1}{5}\sum_{m=-2}^{2}{\nu_m}^2 - {\nu_t}^2\left(1.9 + 1.2\left(\frac{g}{8\pi^2{\nu_t}^2 R_0}\right)^2\right)\right\}, \tag{3}$$

$$R_0 = \left(\frac{3M}{4\rho\pi}\right)^{1/3}, \tag{4}$$

where ν_m is the fundamental surface oscillation frequency of each m mode, ν_t is the translational frequency of the center of gravity, R_0 is the radius of the spherical sample and g is the gravitational constant. In this study, top-view images of the oscillating sample droplet were recorded by the high-speed camera at 500 frames/s for 30 s at each temperature.

Table 1
Chemical compositions of starting materials and after heat capacity measurement.

Chemical composition of starting materials (mass%)											
Sample	Co	Cr	Mo	Ni	Fe	Si	Mn	C	N	O	S
LC	Bal.	27.4	6.09	0.02	0.05	0.59	0.600	0.053	0.1900	0.0211	0.0006
HC	Bal.	27.1	6.01	0.02	0.09	0.54	0.598	0.251	0.1913	0.0120	0.0004

Chemical composition after heat capacity measurement (mass%)											
Sample	Co	Cr	Mo	Ni	Fe	Si	Mn	C	N	O	S
LC	Bal.	27.1–27.3	6.19–6.20	0.02	0.05–0.06	0.54–0.55	<0.001–0.051	0.004–0.031	<0.0001	0.0044–0.0026	0.0007–0.0019
HC	Bal.	26.9–27.8	6.24–6.25	0.02	0.09–0.12	0.53	<0.001–0.052	0.218–0.227	<0.0001–0.0012	<0.0001–0.0059	0.0005–0.0009

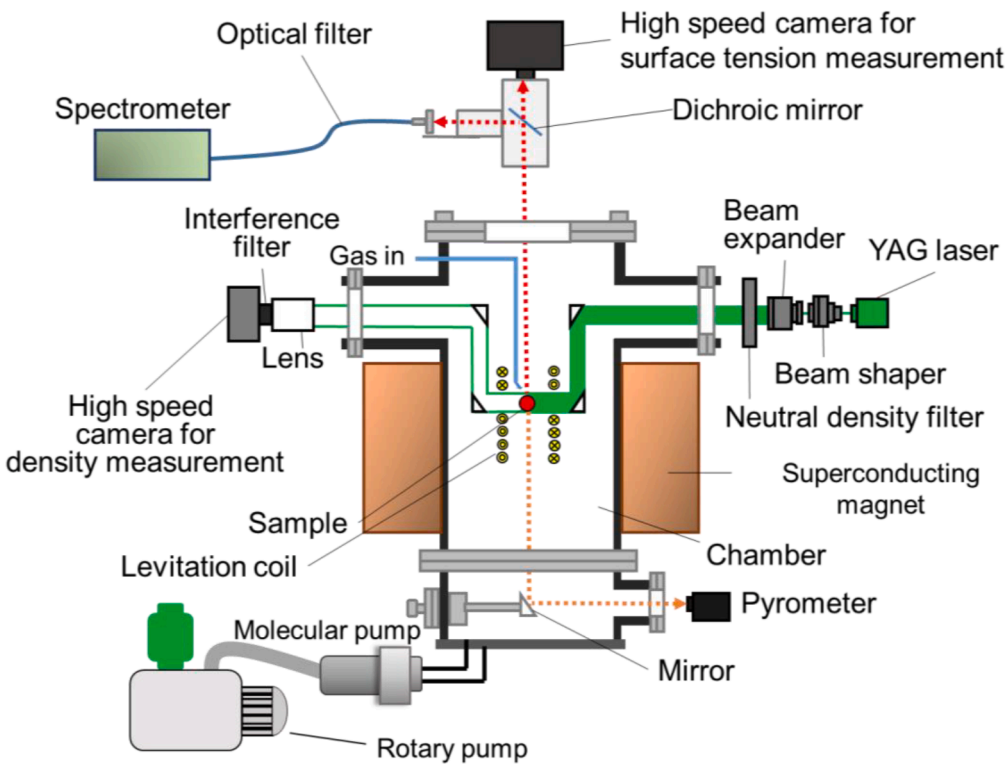

Fig. 1. Schematic illustration of experimental equipment.

The frequencies of the fundamental surface oscillation of the $m = 0, \pm 1$ and $\pm$ 2 modes and ν_t were obtained by fast Fourier transformation (FFT) of the radii and the projection area of the droplet was extracted from consecutive images.

2.2.3. Normal spectral emissivity measurement

The normal spectral emissivity (ε) is expressed as:

$$\varepsilon(\lambda,\ T) = \frac{R_s(\lambda,\ T)}{R_B(\lambda,\ T)}, \tag{5}$$

$$R_s(\lambda,\ T) = D(\lambda)\cdot X(\lambda,\ T), \tag{6}$$

where $R_s(\lambda,\ T)$, $R_B(\lambda,\ T)$, $X(\lambda,\ T)$ and $D(\lambda)$ are the normal spectral radiance that is emitted from the sample, the normal spectral radiance that is emitted from the black body, the calibration coefficient and the spectral output count, respectively. In a previous report [19], the relationship between $R_s(\lambda,\ T)$ and $X(\lambda,\ T)$ was investigated by using a quasi-blackbody, and $D(\lambda)$ was obtained. The $X(\lambda,\ T)$ from the top of the sample droplet was measured by a multichannel spectrometer (USB2000, Ocean Optics Inc., FL, USA). The $R_B(\lambda,\ T)$ was calculated by using Planck's law. The sample was levitated electromagnetically at a static magnetic field of 4 T.

2.2.4. Heat capacity measurement

The sample heat capacity at constant pressure (C_p) was measured by laser modulation calorimetry [16]. A modulated laser at a power (P_0) with an angular frequency (ω) heated the levitated sample at the top part and the temperature response was observed at the bottom of the sample with a pyrometer. The temperature amplitude (ΔT_{ac}) and phase shift ($\Delta\phi$) between the laser intensity and temperature response are expressed as follows:

$$C_p = \frac{\alpha S_h A P_0}{\omega M \Delta T_{ac}}\left(1 + \frac{1}{\omega^2 \tau_r{}^2} + \omega^2 \tau_c{}^2\right)^{-\frac{1}{2}}, \tag{7}$$

$$\Delta\phi = \arccos\left\{\frac{\tau_c}{\omega}\left(\frac{1}{\tau_c \tau_r} - \omega^2\right)\left(1 + \frac{1}{\omega^2 \tau_r{}^2} + \omega^2 \tau_c{}^2\right)^{-\frac{1}{2}}\right\}, \tag{8}$$

where α is the laser absorptivity, S_h is the area ratio of the laser-irradiated part of the levitated sample, A is the superficial area of the levitated sample, and τ_r and τ_c are the external time and internal thermal relaxation times, respectively. α was estimated from the normal spectral emissivities according to Kirchhoff's law.

Quasi-adiabatic conditions, i.e., $\omega^2 \tau_r{}^2 >> 1 >> \omega^2 \tau_c{}^2$, were achieved by a proper choice of modulation frequency and static magnetic field. The heat relaxation times were determined by curve fitting Eq. (8) to a measured relation between $\Delta\phi$ and ω. In the C_p measurements, a relatively low magnetic field intensity between 1 and 4 T was applied to reduce the internal relaxation time by a remaining convection flow in the droplet, which contributed to satisfy the adiabatic condition. The value of C_p was determined from τ_c, τ_r and ΔT_{ac} using Eq. (7).

2.2.5. Thermal conductivity measurement

The thermal conductivity (κ) was also measured by laser modulation calorimetry [16]. The unsteady state heat conduction equation for a spherical droplet is expressed as:

$$\rho C_p \frac{\partial T}{\partial t} = \kappa\left\{\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial T}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial T}{\partial \theta}\right)\right\} + Q(r,\ \theta), \tag{9}$$

where $Q(r, \theta)$ is the heat that is generated by the induction current, and r and θ denote the radial distance and polar angle, respectively. Eq. (9) was solved with the following boundary conditions.

At the droplet surface irradiated by the laser:

$$-\kappa \frac{\partial T}{\partial n} = \sigma_{SB}\varepsilon\left(T^4 - T_a{}^4\right) - \alpha I_0(t)\exp\left(-\frac{2r^2 \sin^2\theta}{(r_{laser})^2}\right)(-\boldsymbol{n}\cdot\boldsymbol{e}_{laser}); \tag{10}$$

at the droplet surface in the absence of the light source:

$$-\kappa\frac{\partial T}{\partial n} = \sigma_{SB}\varepsilon\left(T^4 - T_a{}^4\right); \tag{11}$$

at the centerline:

$$-\kappa\frac{\partial T}{\partial \theta} = 0, \tag{12}$$

where n and $\boldsymbol{n}$ are the normal distance from the droplet surface and unit normal vector at the droplet surface, respectively. T_a is the ambient temperature, σ_{SB} is the Stefan–Boltzmann constant, r_{laser} is the laser beam radius, $\boldsymbol{e}_{laser}$ is the unit vector pointing in the incident direction of the laser beam and $I_0(t)$ is the intensity of the laser beam at the centerline.

The values of $\Delta\phi$ were obtained by solving Eqs. (9)–(12) as follows:

$$\Delta\phi = \tan^{-1}\left(\frac{\Delta T_{ac}{}^{out}}{\Delta T_{ac}{}^{in}}\right), \tag{13}$$

where $\Delta T_{ac}{}^{in}$ and $\Delta T_{ac}{}^{out}$ are the in-phase and out-of-phase component of the temperature amplitude, respectively. κ is determined by curve fitting Eq. (13) to the experimentally obtained relationship between $\Delta\phi$ and ω. In the thermal conductivity measurement, a relatively high magnetic field intensity was applied to suppress the convection flow inside the droplet, which is described in Section 3.7.

3. Results

3.1. Liquidus temperature

The DSC curves of the LC and HC samples during heating are shown in Fig. 2. The DSC curves during heating were used to determine the liquid temperature to avoid undercooling. The liquidus temperatures of the LC and HC samples were determined at the endset temperatures by intersection of the extrapolated base line and tangential line fit to the heat flow curves for the second and third runs. The liquidus temperatures of the LC and HC samples were determined to be 1691 K and 1681 K, respectively. The uncertainty in determination of the liquidus temperature was ~1% and was determined by the DSC resolution accuracy.

3.2. Chemical composition change after heat capacity measurements

Table 1 shows the chemical composition of the LC and HC alloys after the heat capacity measurements. A comparison with the starting materials in Table 1 shows that the chemical contents of Mn, N and O decreased because of evaporation during the measurement. The C content also decreased; however, the reduction was small and the composition of the C of HC alloys remained higher compared with the LC alloys for the samples after the measurement.

3.3. Density

Fig. 3 shows the temperature dependence of the densities of the LC (ρ_{LC}) and HC (ρ_{HC}) melts, which were expressed as a linear function of absolute temperature (T), including the supercooled temperatures below:

$$\rho_{LC} \,/\, \mathrm{kg{\cdot}m^{-3}} = -0.899(T - 1691) + 7573 \quad (1559 \le T \,/\, \mathrm{K} \le 1874\ \mathrm{K}), \tag{14}$$

$$\rho_{HC} \,/\, \mathrm{kg{\cdot}m^{-3}} = -1.073(T - 1681) + 7586 \quad (1529 \le T \,/\, \mathrm{K} \le 1837\ \mathrm{K}). \tag{15}$$

The error bars show the expanded uncertainty ($k = 2$), which is described in Section 4.1.1. The ρ_{LC} and ρ_{HC} were almost the same within

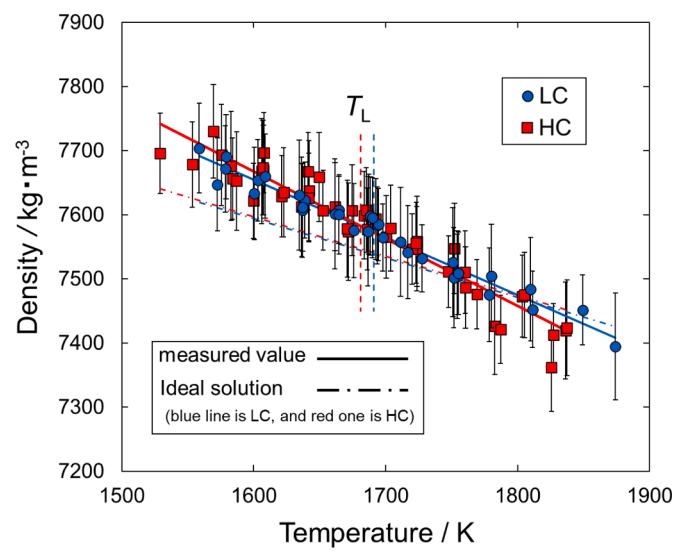

Fig. 3. Temperature dependence of densities of LC and HC melts. T_L is liquidus temperature. Error bars show the expanded uncertainty ($k = 2$).

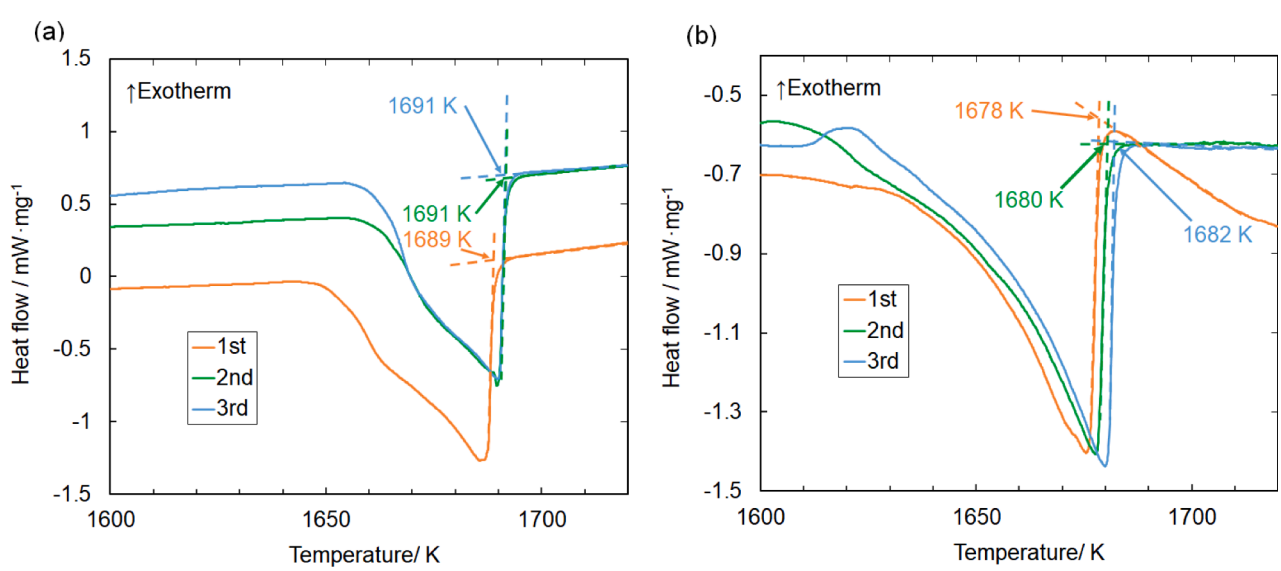

Fig. 2. Example of temperature dependence of heat flow during heating using DSC: (a) LC sample, (b) HC sample.

experimental uncertainty. In Fig. 3, the ρ of the LC and HC melts as calculated by an ideal solution model is also presented by dash-dot-dash lines. The ideal solution model is explained in Section 4.2.

3.4. Surface tension

Fig. 4 shows typical oscillation spectra of the Area and the R^+ and R^- of the HC melt at 1800 K obtained by FFT. Five peaks appeared in the spectra as shown in Fig. 4. Area is the projection area of the droplet image. R^+ and R^- are the sum and difference of the radius along the x axis (R_x) and y axis (R_y), respectively, of the droplet image with an origin at the center of gravity of the droplet. The five peaks were assigned by identification rules shown in Table 2. Fig. 5 shows the temperature dependence of the surface tension of the LC (σ_{LC}) and HC (σ_{HC}) melts. The error bars show the expanded uncertainty, which is discussed in Section 4.1.2. The σ_{LC} and σ_{HC} have a positive temperature dependence at lower temperatures and the temperature dependence became negative at a higher temperature. This temperature dependence is commonly termed the "boomerang shape", and a similar behavior has been observed in the surface tensions of other melts [22–26]. This behavior tends to be caused by surface segregation of the alloy components with a low surface tension or adsorption–desorption equilibrium of surface-active elements, such as oxygen or sulfur on the melt surface. As presented in Table 1, the LC and HC samples initially contained 211 and 120 mass ppm oxygen, respectively, and therefore, the σ_{LC} and σ_{HC} could be influenced by oxygen absorption on the melt surface.

The temperature dependence of the σ_{LC} and σ_{HC} was almost the same, therefore, the surface tension of the CCM melts (σ_{CCM}) was determined by both data sets and compiled as follows:

$$\sigma_{CCM} \,/\, \mathrm{mN{\cdot}m^{-1}} = -3.44 \times 10^{-6} T^3 + 1.67 \times 10^{-2} T^2 - 26.4T + 15313 \quad (1589 \le T \,/\, \mathrm{K} \le 1986\ \mathrm{K}). \tag{16}$$

3.5. Normal spectral emissivity

Fig. 6 shows the temperature dependence of the normal spectral emissivity of the LC (ε_{LC}) and HC (ε_{HC}) melts at 940 nm. The error bars show the expanded uncertainty, which is discussed in Section 4.1.3. ε_{LC} and ε_{HC} showed no temperature dependence and were almost the same. The ε_{LC} and ε_{HC} at 940 nm were determined as:

$$\varepsilon_{LC} = 0.30 \pm 0.02 \quad (1583 \le T \,/\, \mathrm{K} \le 1796\ \mathrm{K}), \tag{17}$$

$$\varepsilon_{HC} = 0.30 \pm 0.02 \quad (1581 \le T \,/\, \mathrm{K} \le 1828\ \mathrm{K}). \tag{18}$$

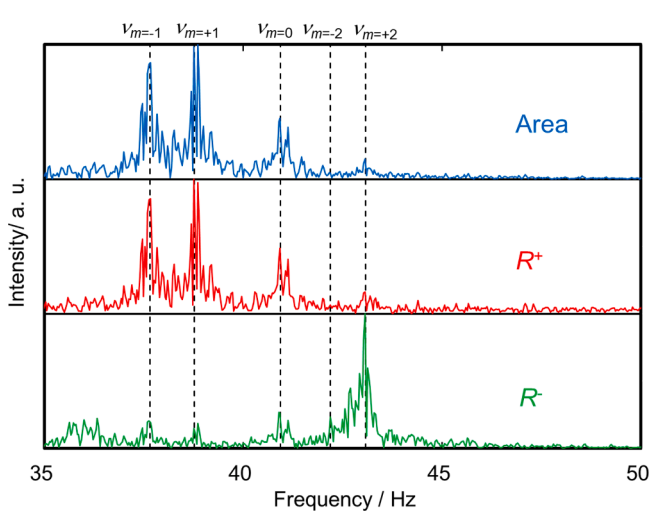

Fig. 4. Frequency spectra of surface oscillation of area, R^+ and R^- of HC melt at 1800 K.

Table 2
Identification rule for surface oscillation.

	$m = 0$	$m = \pm1$	$m = \pm2$
Area	Yes	Yes	No
R^+	Yes	Yes	No
R^-	No	Yes	Yes

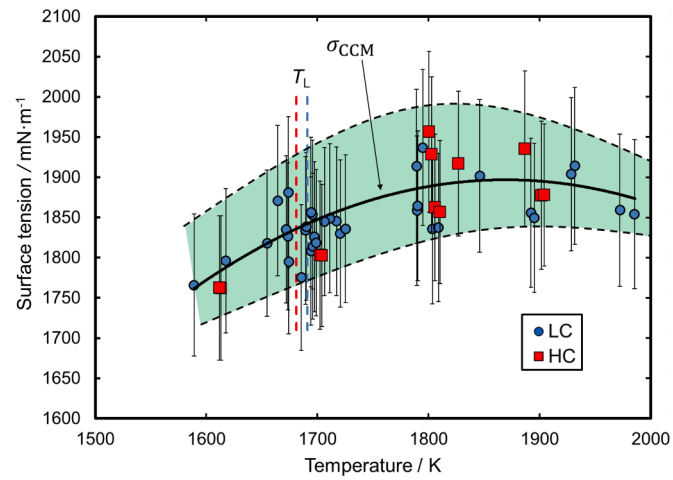

Fig. 5. Temperature dependence of surface tension of LC and HC melts. Error bars indicate expanded uncertainty ($k = 2$). Dashed lines show liquidus temperatures T_L of LC and HC melts. σ_{CCM} shows complied surface tension of CCM melts.

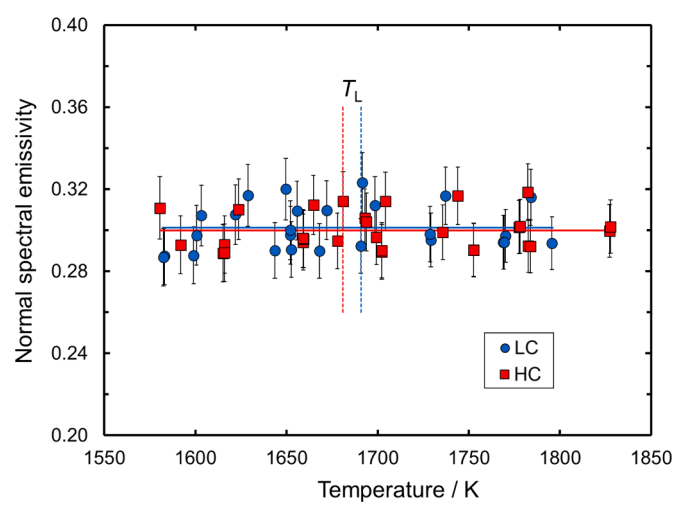

Fig. 6. Temperature dependence of normal spectral emissivities of LC and HC melts at 940 nm. Dashed lines show liquidus temperatures T_L of LC and HC melts. Error bars indicate expanded uncertainty ($k = 2$).

3.6. Heat capacity at constant pressure

Fig. 7 shows a typical laser power irradiated on the sample top and the temperature response measured at the sample bottom during the laser modulation calorimetry of the LC melt at an angular frequency (ω) of 0.63 rad·s^{-1} under a static magnetic field of 4 T. The phase shift ($\Delta\phi$) between the laser power and the sample temperature and temperature amplitude (ΔT_{ac}) are presented in Fig. 7.

Fig. 8 shows the $\Delta\phi$ and $\omega\Delta T_{ac}$ as a function of angular frequency (ω) at a static magnetic field of 4 T at 1711 K. The value of $\omega\Delta T_{ac}$ was maximum at a ω of ~0.50 rad·s^{-1}, which satisfied the adiabatic

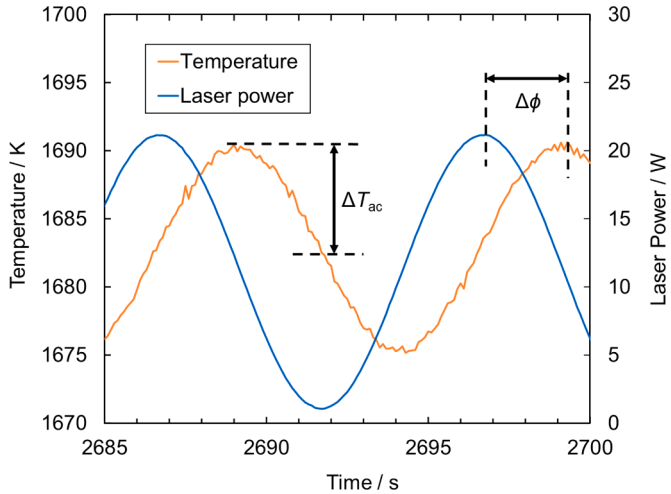

Fig. 7. Laser power (right-hand ordinate) and temperature response (left-hand ordinate) of LC melt during laser-modulated calorimetry with angular frequency of 0.63 rad·s^{-1} under static magnetic field of 4 T.

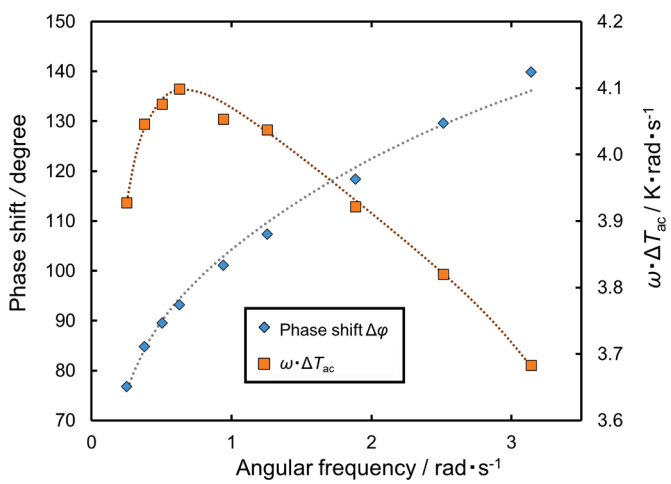

Fig. 8. Phase shift (left-hand ordinate) and $\omega{\cdot}\Delta T_{ac}$ (left-hand ordinate) as a function of modulation frequency of LC melt at 1711 K and 4 T.

condition that corresponded to $\Delta\phi = 90°$. The τ_r and τ_c were evaluated as 0.35 s and 11.38 s, respectively, by fitting the experimental phase shift to Eq. (8). The heat capacity (C_p) of LC was obtained by Eq. (7) by using a maximum of $\omega\Delta T_{ac}$.

The specific heat capacities of the LC ($C_{P,\ LC}$) and HC ($C_{P,\ HC}$) melts are presented in Fig. 9. The $C_{P,\ LC}$ and $C_{P,\ HC}$ were almost the same within experimental uncertainty. The values had a negligible temperature dependence and were determined as:

$$C_{P,\ LC}\ /\ \mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}} = 837\ \pm\ 17\quad (1593 \le T\ /\ \mathrm{K} \le 1863\ \mathrm{K}), \tag{19}$$

$$C_{P,\ HC}\ /\ \mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}} = 840\ \pm\ 15\quad (1589 \le T\ /\ \mathrm{K} \le 1866\ \mathrm{K}). \tag{20}$$

The $C_{P,\ LC}$ and $C_{P,\ HC}$ as calculated by an ideal solution model were plotted in Fig. 9, as discussed in Section 4.2.

3.7. Thermal conductivity

Fig. 10 shows the static magnetic field dependence of the apparent thermal conductivities of the LC and HC melts at 1800 K. The apparent thermal conductivities of the LC and HC melts decreased with an increase in static magnetic field and the values approached a constant value at 9 T or above within experimental uncertainty. This result shows

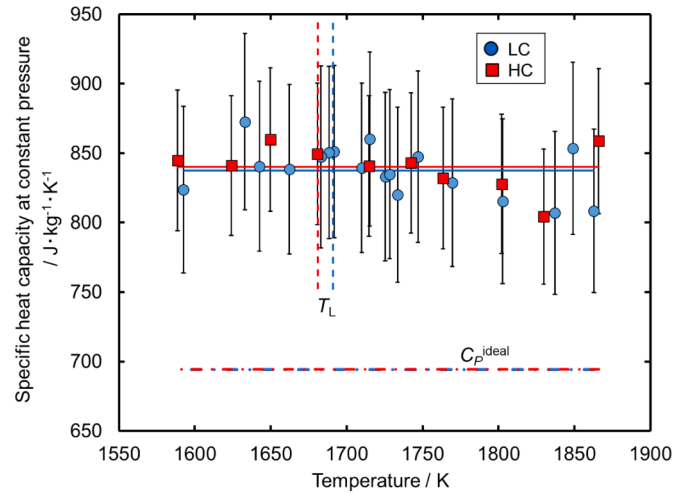

Fig. 9. Temperature dependence of specific heat capacities at constant pressure of LC and HC melts. Dashed lines show liquidus temperatures T_L of LC and HC melts. Error bars indicate expanded uncertainty ($k = 2$). Dotted and dashed lines show heat capacity of ideal solution model of LC (blue line) and HC (red line) melts.

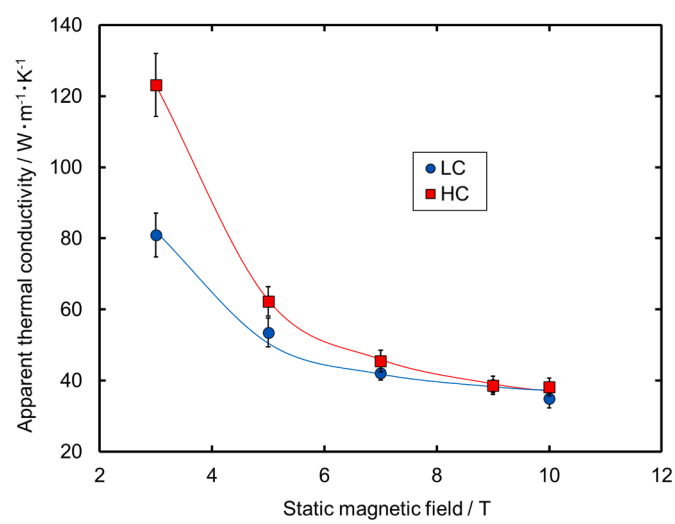

Fig. 10. Static magnetic field dependence of apparent thermal conductivities of LC and HC melts at 1800 K. Error bars indicate expanded uncertainty ($k = 2$).

that the convection flow of LC and HC was suppressed sufficiently to enable a true thermal conductivity measurement above 9 T. Based on these results, the thermal conductivity measurements of the LC and HC melts were carried out at 10 T.

Fig. 11 shows the temperature dependence of the thermal conductivity of LC (κ_{LC}) and HC (κ_{HC}).

In this study, the thermal conductivities were measured above 1749 K because stable levitation of LC and HC melts was difficult at a high magnetic field of 10 T at lower temperatures close to the liquidus temperature because of the large magnetic force. The static magnetic field is useful to suppress surface oscillation and translational motion of the levitated droplet. However, when performing measurements on a sample with a large magnetic susceptibility, such as Co, the levitated sample receives a large magnetic force generated by the magnetic gradient [27, 28]. A magnetic field strength greater than 9 T was required for the thermal conductivity measurement to suppress the internal convection, which made the levitation of CCM unstable.

κ_{LC} and κ_{HC} increased with an increase in temperature, thus, κ_{LC} and κ_{HC} were determined by least-squares approximation as:

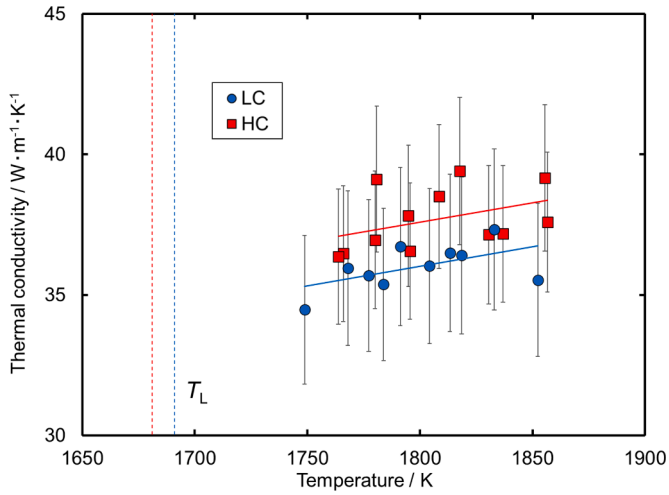

Fig. 11. Temperature dependence of thermal conductivities of LC and HC melts in a static magnetic field of 10 T. Dashed lines show liquidus temperatures T_L of LC and HC melts. Error bars indicate expanded uncertainty ($k = 2$).

$$\kappa_{LC} \,/\, \mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}} = 0.014(T - 1691) + 34.7 \quad (1749 \le T \,/\, \mathrm{K} \le 1852\ \mathrm{K}), \tag{21}$$

$$\kappa_{HC} \,/\, \mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}} = 0.014(T - 1681) + 36.5 \quad (1764 \le T \,/\, \mathrm{K} \le 1857\ \mathrm{K}). \tag{22}$$

κ_{HC} was larger by 5% than κ_{LC}; however, considering the expanded uncertainty (6.0–7.7%) in the thermal conductivity measurement as explained in Section 4.1.5, the difference between κ_{LC} and κ_{HC} is not very meaningful.

4. Discussion

4.1. Uncertainty analysis

In this study, the measurement uncertainty was conducted based on the Guide to the Expression of Uncertainty in Measurement (GUM) [29].

4.1.1. Uncertainty analysis of the density measurement

The uncertainty in density measurement was evaluated by using the following equation:

$$u^2(\rho) = \left(\frac{\partial\rho}{\partial V}\right)^2 u^2(V) + \left(\frac{\partial\rho}{\partial M}\right)^2 u^2(M), \tag{23}$$

where $u(\rho)$ is the combined standard uncertainty in the density measurement. $u(V)$ is the uncertainty from the standard deviation of values of volume evaluated from 3000 droplet images. $u(M)$ is the uncertainty in mass that was caused by sample evaporation. By way of example, the uncertainty analysis for the HC melt at 1671 K is presented in Table 3. Table 3 shows that the dominant uncertainty was $u(M)$, and the expanded uncertainty ($U = 2u(\rho)$ (95.45% confidence)) was 76 kg·m^{-3}, which corresponded to 1.0% of the density value (7578 kg·m^{-3}). In this study, U of the LC and HC melts ranged between 0.7% and 1.1%. The accuracy of the pyrometer was 0.5%. The contribution of the pyrometer accuracy to the density measurement uncertainty was evaluated to be 0.1%, which was considerably lower than other factors.

4.1.2. Uncertainty analysis of surface tension measurement

The uncertainty of the surface tension is expressed as:

$$u^2(\sigma) = \left(\frac{\partial\sigma}{\partial M}\right)^2 u^2(M) + \left(\frac{\partial\sigma}{\partial \nu_R}\right)^2 u^2(\nu_R), \tag{24}$$

Table 3
Uncertainty evaluation in density measurements of HC melt at 1671 K.

Component	Standard uncertainty	Sensitivity coefficient	Contribution
Uncertainty of volume: u (V)	2.3×10^{-10} m^3	$\frac{\partial\rho}{\partial V} = -6.1 \times 10^{10}$ kg·m^{-6}	−14 kg·m^{-3}
Uncertainty of mass loss due to evaporation: u (M)	4.4×10^{-6} kg	$\frac{\partial\rho}{\partial M} = 8.0 \times 10^{6}$ m^{-3}	35 kg·m^{-3}
Combined standard uncertainty $u(\rho)$			38 kg·m^{-3}
Expanded uncertainty $U = 2u(\rho)$ (95.45% confidence)			76 kg·m^{-3} (1.0%)

$\rho = 7578$ kg·m^{-3} for HC melt at 1671 K.

where $u(\sigma)$ is the combined standard uncertainty of the surface tension. $u(\nu_R)$ is the uncertainty of surface oscillation frequency. $u(\nu_R)$ was evaluated from a resolution of frequency in FFT analysis. As an example, the uncertainty evaluation of σ of the HC melt at 1800 K is listed in Table 4. The dominant contribution in the uncertainty was $u(\nu_R)$ and $U = 2u(\sigma)$ was 100 mN·m^{-1}, which corresponds to 5.1%. In this study, U for all measurements of LC and HC melts ranged from 5.0% to 5.1% and 4.5% to 5.1%, respectively. The contribution of the pyrometer accuracy to the surface tension measurement uncertainty was evaluated to be 0.4%, which was considerably lower than other factors.

4.1.3. Uncertainty analysis of normal spectral emissivity measurement

The uncertainty of the normal spectral emissivity is expressed as:

$$u^2(\varepsilon) = \left(\frac{\partial\varepsilon}{\partial D}\right)^2 u^2(D) + \left(\frac{\partial\varepsilon}{\partial T}\right)^2 u^2(T), \tag{25}$$

where $u(\varepsilon)$ is the combined standard uncertainty of the normal spectral emissivity. The factor $u(D)$ is the uncertainty of calibration coefficient D of Eq. (6), which was evaluated from the uncertainty of the slope between R_s and X [21]. The factor $u(T)$ is the uncertainty of the pyrometer, which was contributed to the uncertainty of R_B through Planck's law. For instance, the uncertainty evaluation of the emissivity of the LC melt at 1698 K is presented in Table 5. The dominant contribution in the uncertainty was $u(T)$ and the expanded uncertainty ($U = 2u(\varepsilon)$) was 0.014, which corresponded to 4.5% of the emissivity. In this study, U for all measurements of the LC and HC melts ranged from 4.3% to 4.9%.

4.1.4. Uncertainty analysis of heat capacity

The uncertainty of the heat capacity is given by:

$$u^2(C_P) = \left(\frac{\partial C_P}{\partial\varepsilon}\right)^2 u^2(\varepsilon) + \left(\frac{\partial C_P}{\partial\Delta T_{ac}}\right)^2 u^2(\Delta T_{ac}) + \left(\frac{\partial C_P}{\partial M}\right)^2 u^2(M), \tag{26}$$

where $u(C_P)$ is the combined standard uncertainty of the heat capacity. The factors $u(\varepsilon)$ and $u(\Delta T_{ac})$ are the uncertainties of the normal spectral emissivity, and temperature amplitude, respectively. For example, the

Table 4
Uncertainty evaluation in surface tension measurements for HC melt at 1800 K.

Component	Standard uncertainty	Sensitivity coefficient	Contribution
Uncertainty of mass loss due to evaporation: $u(M)$	2.5×10^{-6} kg	$\frac{\partial\sigma}{\partial M} = 1.84 \times 10^{3}$ N·m^{-1}	4.7 mN·m^{-1}
Resolution of frequency: u (ν_R)	0.5 Hz	$\frac{\partial\sigma}{\partial\nu_R} = 9.9 \times 10^{-2}$ N·m^{-1}·s	50 mN·m^{-1}
Combined standard uncertainty $u(\sigma)$			50 mN·m^{-1}
Expanded uncertainty $U = 2u(\sigma)$ (95.45% confidence)			100 mN·m^{-1}

$\sigma = 1965$ mN·m^{-1} for HC melt at 1800 K.

Table 5
Uncertainty evaluation in normal spectral emissivity measurements of LC melt at 1698 K.

Component	Standard uncertainty	Sensitivity coefficient	Contribution
Uncertainty of calibration coefficient (D): $u(D)$	2.18×10^{-3} W·m^{-2}·μm^{-1}	$\frac{\partial\varepsilon}{\partial D} = 2.51 \times 10^{-1}$ W^{-1}·m^2·μm	5.47×10^{-4}
Uncertainty of pyrometer: $u(T)$	4.25 K	$\frac{\partial\varepsilon}{\partial T} = -1.66 \times 10^{-3}$ K^{-1}	-7.03×10^{-3}
Combined standard uncertainty $u(\varepsilon)$			0.007
Expanded uncertainty $U = 2u(\varepsilon)$ (95.45% confidence)			0.014

$\varepsilon = 0.312$ for LC melt at 1698 K.

uncertainty evaluation of the heat capacity measurement in the LC melt at 1662 K is listed in Table 6. The dominant contribution in the uncertainty was $u(\varepsilon)$ and the expanded uncertainty ($U = 2u(C_P)$) was 62.0 J·kg^{-1}·K^{-1}, which corresponded to 7.4% of the heat capacity. In this study, U for all measurements of the LC and HC melts ranged from 7.4% to 7.9% and 6.1% to 6.2%, respectively.

4.1.5. Uncertainty analysis of thermal conductivity measurement

The uncertainty of the thermal conductivity $u(\kappa)$ was evaluated by:

$$u^2(\kappa) = \left(\frac{\partial\kappa}{\partial\Delta\varphi}\right)^2 u^2(\Delta\phi) + \left(\frac{\partial\kappa}{\partial\rho}\right)^2 u^2(\rho) + \left(\frac{\partial\kappa}{\partial C_P}\right)^2 u^2(C_P). \tag{27}$$

where $u(\Delta\phi)$ is the uncertainty of the phase shift. Table 7 presents a list of uncertainty components in the thermal conductivity measurement. The dominant contribution in the uncertainty was $u(C_P)$ and the expanded uncertainty ($U = 2u(\kappa)$) was 2.64 W·m^{-1}·K^{-1}, which corresponded to 7.3% of the thermal conductivity. $U = 2u(\kappa)$ for all measurements for the LC and HC melts ranged from 7.2% to 7.7% and 6.0% to 6.1%, respectively. The contribution of the pyrometer accuracy to the surface tension measurement uncertainty was evaluated to be 0.3%, which was considerably lower than other factors.

4.2. Density and heat capacity of ideal solution model

The densities and heat capacities of the LC and HC melts were calculated by assuming that an ideal solution consisted of Co, Cr and Mo ternary systems and by ignoring other minor elements. The data used in this calculation are presented in Table 8. Here, the data for pure Cr and Mo melts were extrapolated to the present experimental temperature range.

The density of the ideal solution model (ρ^{ideal}) is expressed as:

$$\rho^{\text{ideal}} = \sum_{i=0}^{N} \frac{x_i \cdot M_i}{V^{\text{ideal}}}, \tag{28}$$

Table 6
Uncertainty evaluation in heat capacity measurements for LC melt at 1662 K.

Component	Standard uncertainty	Sensitivity coefficient	Contribution
Uncertainty of normal spectral emissivity: $u(\varepsilon)$	0.01	$\frac{\partial C_P}{\partial\varepsilon} = 2.80 \times 10^3$ J·kg^{-1}·K^{-1}	30.9 J·kg^{-1}·K^{-1}
Uncertainty of temperature amplitude: $u(\Delta T_{ac})$	0.02 K	$\frac{\partial C_P}{\partial\Delta T_{ac}} = -107$ J·kg^{-1}·K^{-2}	-2.52 J·kg^{-1}·K^{-1}
Uncertainty of mass loss due to evaporation: $u(M)$	7.10×10^{-6} kg	$\frac{\partial C_P}{\partial M} = 118 \times 10^3$ J·kg^{-2}·K^{-1}	8.39×10^{-3} J·kg^{-1}·K^{-1}
Combined standard uncertainty $u(C_P)$			31.0 J·kg^{-1}·K^{-1}
Expanded uncertainty $U = 2u(C_P)$ (95.45% confidence)			62.0 J·kg^{-1}·K^{-1}

$C_P = 838$ J·mol^{-1}·K^{-1} for LC melt at 1662 K.

Table 7
Uncertainty evaluation in thermal conductivity measurements of LC melt at 1749 K.

Component	Standard uncertainty	Sensitivity coefficient	Contribution
Uncertainty of phase shift: $u(\Delta\phi)$	1 °	$\frac{\partial\kappa}{\partial(\Delta\varphi)} = 0.20$ W·m^{-1}·K^{-1}·°$^{-1}$	0.20 W·m^{-1}·K^{-1}
Uncertainty of density: $u(\rho)$	52.0 kg·m^{-3}	$\frac{\partial\kappa}{\partial\rho} = 0.01$ W·m^2·K^{-1}·kg^{-1}	0.24 W·m^{-1}·K^{-1}
Uncertainty of heat capacity: $u(C_P)$	31.2 J·kg^{-1}·K^{-1}	$\frac{\partial\kappa}{\partial C_P} = 0.41$ s^{-1}·m^{-1}·kg^{-1}	1.28 W·m^{-1}·K^{-1}
Combined standard uncertainty $u(\kappa)$			1.32 W·m^{-1}·K^{-1}
Expanded uncertainty $U = 2u(\kappa)$ (95.45% confidence)			2.64 W·m^{-1}·K^{-1}

$\kappa = 34.5$ W·m^{-1}·K^{-1} for LC melt at 1749 K.

Table 8
Values used to calculate ideal solution of LC and HC melts.

Values	LC	HC
Mole fraction	x_{Co}=0.647, x_{Cr}=0.313, x_{Mo}=0.040	x_{Co}=0.650, x_{Cr}=0.310, x_{Mo}=0.040
Molar mass / g·mol^{-1}	58.30	58.29
Density of ideal solution / kg·m^{-3}	−0.61(T-1691)+7538	−0.61(T-1681)+7547
Specific heat capacity at constant pressure / J·kg^{-1}·K^{-1}	694.2	694.5
Molar heat capacity at constant pressure / J·mol^{-1}·K^{-1}	48.0	47.8
Excess heat capacity / J·mol^{-1}·K^{-1}	7.52 (1593 ≤ T/K ≤ 1863 K)	7.36 (1589 ≤ T/K ≤ 1866 K)
The density of Co melt [28] / kg·m^{-1}	−0.54 (T-1768)+7783 (T = 1757 ≤ T/K ≤ 2001 K)	
The density of Cr melt [30] / kg·m^{-1}	−0.72 (T-2130)+7830 (T = 2163 ≤ T/K ≤ 2503 K)	
The density of Mo melt [31] / kg·m^{-1}	−0.60 (T- 2896)+9100 (T = 2500 ≤ T/K ≤ 3000 K)	
The molar heat capacity of Co melt [28] / J·mol^{-1}·K^{-1}	41.2 (T = 1779 ≤ T/K ≤ 2023 K)	
The molar heat capacity of Cr melt [32] / J·mol^{-1}·K^{-1}	39.33 (T = 2130 ≤ T/K ≤ 2900 K)	
The molar heat capacity of Mo melt [32] / J·mol^{-1}·K^{-1}	37.656 (T = 2896 ≤ T/K ≤ 4951 K)	

$$V^{\text{ideal}} = \sum_{i=0}^{N} \frac{x_i \cdot M_i}{\rho_i}, \tag{29}$$

where V^{ideal} is the molar volume of the ideal solution model and x_i, M_i and ρ_i are the mole fraction, molar mass and density of component i, respectively. The ρ^{ideal} of the LC and HC melts are shown in Fig. 3 and Table 8. The ρ^{ideal} of the LC and HC melts agrees with the experimental ρ_{LC} and ρ_{HC} within experimental uncertainty.

The molar heat capacity at a constant pressure of an ideal solution ($C_{P,\ m}{}^{\text{ideal}}$) and the excess heat capacity ($C_P{}^{E}$) are given by:

$$C_{P,\ m}{}^{\text{ideal}} = \sum_{i=0}^{N} x_i \cdot C_{P,m,\ i}, \tag{30}$$

$$C_P{}^{E} = C_{P,\ m} - C_{P,\ m}{}^{\text{ideal}} \tag{31}$$

$$= \frac{\partial H^{E}}{\partial T}. \tag{32}$$

$C_{P,\ m,\ i}$ is the molar heat capacity of component *i*, $C_{P,\ m}$ is the molar heat capacity of real alloys and H^E is the excess enthalpy. The specific heat capacity of an ideal solution (C_P^{ideal}) of the LC and HC melts was obtained by conversion from $C_{P,\ m}^{ideal}$, and they were presented by dash-dot-dash lines in Fig. 9. Fig. 9 shows that the experimental C_P was 20% larger than the C_P^{ideal}. Thus, the C_P^E of the LC and HC melts were positive and therefore, the temperature dependence of H^E was also positive. The Lupis–Elliott rule [33] explains that excess properties approach zero with an increase in temperature. From these results and the Lupis–Elliott rule, the H^E of the LC and HC melts should be negative in the measurement temperature region and approaches zero with increasing temperature.

5. Conclusions

The liquidus temperatures of the Co–Cr–Mo alloys with different initial carbon contents (LC (0.053 mass%) and HC (0.251 mass%)) were determined as 1691 K and 1681 K, respectively, by DSC. The density, surface tension, normal spectral emissivity, heat capacity and thermal conductivity of the LC and HC melts were measured by EML in a static magnetic field. No difference was found in the thermophysical properties of the LC and HC melts within experimental uncertainty. An uncertainty analysis was conducted for the thermophysical property data. An ideal solution model reproduced the density of the CCM melts within experimental uncertainty but we obtained large positive excess heat capacities of the CCM melts.

CRediT authorship contribution statement

Manabu Watanabe: Conceptualization, Investigation, Writing – original draft. **Yuki Takahashi:** Investigation, Validation. **Sayaka Imaizumi:** Investigation, Validation. **Yufan Zhao:** Conceptualization, Investigation, Methodology. **Masayoshi Adachi:** Conceptualization, Investigation, Methodology, Validation, Supervision. **Makoto Ohtsuka:** Conceptualization, Investigation, Methodology, Validation, Supervision. **Akihiko Chiba:** Conceptualization, Validation, Supervision. **Yuichiro Koizumi:** Conceptualization, Investigation, Methodology, Validation, Supervision, Project administration. **Hiroyuki Fukuyama:** Conceptualization, Investigation, Methodology, Validation, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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